

ICD 1.9/1.10[ab]

“NIRI/GNIRS Science Instrument to On Instrument Wavefront Sensor”

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Chapter 1 Introduction

1–1.0 Description

This document describes the NIRI and GNIRS instrument-to-wavefront-sensor-components interlock system (not to be confused with the Gemini Interlock Bus). These interlocks provide a way for the instrument to inform the wavefront sensor that it should not move its components at this time.

1–1.1 Purpose

This Interface Control Document (ICD) serves the following purposes:

- To describe the commands and parameters that will pass between the Near Infra-Red Imager (NIRI) and its on instrument wavefront sensor (OIWFS) and between the Gemini Near Infra-Red Spectrograph (GNIRS) and its OIWFS.

The intended audience for this document is:

- the NIRI reviewers;
- the GNIRS reviewers and developers.

1–1.2 Scope

This document describes commands which pass between NIRI/GNIRS and their respective OIWFS's.

This document does *not* describe the following topics:

- the NIRI detector and components, (see [18], [11], and [12], summarized in [3]);
- the GNIRS hardware and electronics, (see [18] and (???))
- the OIWFS hardware and electronics, (see [11] and [12], summarized in [3]);
- the NIRI/GNIRS OIWFS detector and components (see [11], [12], and [13], summarized in [3]);

The software design is not described here. For that, see [2] (currently under preparation), and [3] (currently under preparation).

1–1.3 Differences between NIRI and GNIRS

For GNIRS, the commands here affect only the OIWFS. For NIRI, the commands affect the entire instrument.

1–2.0 Related Documents and Drawings

1–2.1 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

Chapter 2 Details of the NIRI/GNIRS to OIWFS Interface

2–1.0 Introduction

2–2.0 Physical System Interfaces

2–2.1 Mechanical Interface

Not Applicable for a software ICD.

2–2.2 Optical Interface

Not Applicable for a software ICD.

2–2.3 Electronic Interface

Not Applicable for this ICD.

2–2.4 Mass/Balance

Not Applicable for a software ICD.

2–2.5 Thermal Interface

Not Applicable for a software ICD.

2–3.0 Software/Control Function Interface (wavefront sensor)

2–3.1 Overview

2–3.1.1 Communications infrastructure

The communications infrastructure is the same as described in [16].

2–3.2 Implementation

This section contains tables summarizing the EPICS records used for the interface. The next sections (Section 2–3.2.1.1 on page 6) contain the details of what those records do.

It is assumed in all cases that the record names given in this document will be prefixed by the string “niri:” or “nirs:” when the database is implemented, except for the SAD database which might be prefixed by “niri::” or “nirs::”.

2–3.2.1 Command (CAD/CAR) records

The columns of the command record tables are as follows:

Description. A brief description of the command.

Attribute. The name of the attribute which the Observatory Control System associates with a particular parameter of a particular command.

CAD record/field. The name of the EPICS CAD record (minus the “niri:” or “nirs:” prefix) together with the field in that CAD record associated with each parameter of the command.

CAR record. The CAR record associated with the command. Note that several CAD records may share the same CAR record.

These commands are executed by asserting the top level WFS apply record.

TABLE 1. NIRI/GNIRS OIWFS CAD/CAR records

Description	Attribute	CAD record:field (“niri:wfs:” +)	CAR record (“niri:wfs:” +)
Configuration interlock	Lock for configuration (string, “LOCK” “UNLOCK”)	lockCfgSel.A	lockC
Observation interlock	Lock for observation (string, “LOCK” “UNLOCK”)	lockObsSel.A	
General interlock	Lock (string, “LOCK” “UNLOCK”)	lockGenSel.A	
Temperature interlock	Lock for cool down / warm up (string, “WARM” “COLD” “CHANGING”)	lockTmpSel.A	
Datum mechanism	Datum mechanisms	datum	datumC

2–3.2.1.1 Detailed Command Description

2–3.2.1.2 lockCfgSel

An instrument may have limitations on which mechanisms which may be moving simultaneously. This interlock is intended to prevent the wavefront sensor components from moving while the instrument is being reconfigured, if it is possible for wavefront sensor components to be damaged. There are two possible states:

“LOCK”. Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

“UNLOCK”. Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2–6.0, “Activity Completion,” p. 11, below.)

This command should be used with caution. Excessive use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding sub-system).

This command is not needed by NIRI.

2-3.2.1.3 lockObsSel

This command is provided to provide a low noise environment for sensitive observations. There are two possible states.

“LOCK”. Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

“UNLOCK”. Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2-6.0, “Activity Completion,” p. 11, below.)

This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).

This command is not needed by NIRI.

2-3.2.1.4 lockGenSel

This command is provided as a “general” command to lock all hardware for any unspecified reason.

There are two possible states:

“LOCK”. Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

“UNLOCK”. Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2-6.0, “Activity Completion,” p. 11, below.)

This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).

This command is not needed by NIRI.

2-3.2.1.5 lockTmpSel

The OIWFS motors (and, possibly, other components) can be damaged if a move command is issued before the system has reached a uniform temperature throughout. Furthermore, the physical parameters of the system (e.g., the datum position) may depend on the temperature. The lockTmpSel command informs the OIWFS about the current temperature of the system. There are three possible states:

“WARM”. Hardware is permitted to move (if no other locks are set). The system is at or near room temperature.

“COLD”. Hardware is permitted to move (if no other locks are set). The system is at or near its operating temperature.

“CHANGING”. Hardware is not permitted to move. The system is currently being cooled down or warmed up.

When the temperature setting is changed, the OIWFS will enter the not-datumed state. (The datum shifts slightly, depending on the temperature of the system.) Upon reboot, the system always begins in a “CHANGING” state, until explicitly set to another state.

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2–6.0, “Activity Completion,” p. 11, below.)

It is always better to be conservative and leave the state of the system “CHANGING”, until it can be guaranteed that the OIWFS mechanisms have reached their steady-state temperature.

This command also sets the value of various internal temperature-dependent parameters within the OIWFS. If this parameter is set incorrectly, mechanisms will move, but may incorrectly report that there is a hardware problem.

This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).

2–3.2.1.6 DATUM

This command will cause all mechanisms to be datumed. Note that the OIWFS does *not* automatically datum itself when a move command is issued.

This command should be used with caution. Use of this lock may interfere with the proper functioning of the OIWFS and will keep the OIWFS from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).

2–3.2.2 Status Information Records

The following tables show the status information provided by the NIRI/GNIRS OIWFS components controller which are of importance to the instrument as a whole.

Unless otherwise noted, the SIR records described here are to be used for status information, and not to be used for real-time control purposes. It is possible that rapid changes in the underlying system may not be reflected in the SIR records.

The tables have the following columns:

SIR record. The part of the name of the EPICS SIR record which follows the “niri:” prefix¹.

Units. The units in which the quantity is stored

Comments. A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

1. Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI database (for example “niri:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:”.

Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.

TABLE 2. NIRI/GNIRS OIWFS status Information

SIR record ("niri:wfs:" +)	FITS keyword	FITS includ ed?	Type	Units	Comments
lockHealth		no	string		Health of subsystem ("GOOD" "WARNING" "BAD"). Intended to be combined with overall instrument Health. This can also be set if there is a hardware or configuration problem with the database. Diagnosis requires the use of the Engineering user interface, and is not considered a function of the Instrument to OIWFS interface.
lockCfg		no	long		Locked for configuration (non-zero = locked). If this is set, the system will be put into a "WARNING" state.
lockObs		no	long		Locked for observation (non-zero = locked). If this is set, the system will be put into a "WARNING" state.
lockGen ^a		no	long		Locked (generic) (non-zero = locked). If this is set, the system will be put into a "WARNING" state.
lockTmp		no	long		Locked for cool down / warm up (non-zero = locked. I.e., lockTmpVal is "CHANGING". If this is set, the system will be put into a "WARNING" state.
lockInit		no	long		Locked for initialization (non-zero = locked)? If this is set, the system will be put into a "WARNING" state.
lockLock		no	long		Combined value (<i>logical or</i>) of lockCfg, lockObs, lockGen, lockTmp, and lockInit.
lockTmpVal		no	string		Current temperature state ("COLD" "WARM" "CHANGING ^b ")
lockMaxMove		no	long		How many motors may be moved at the same time. ^c
lockActive		no	long		How many motors have been commanded to move and are waiting to move or are currently moving ^c .
lockMoving		no	long		How many motors are currently moving ^c .
wfsBeam		no	long		Flag to indicate that the WFS components are allowing light to fall on the on-instrument wavefront sensor. ^d

- The instrument should not trigger any special actions based on the state of the lockGen record, because it is intended that the lockGen should be usable at any time, for any reason, without side effects.
- For safety, when NIRI or the standalone OIWFS is rebooted, the system always restarts in the "CHANGING".
- For NIRI, this number is the total of the NIRI CC motors and NIRI OIWFS motors, *excluding the gimbal (prbx, prby) motors, which may be moved at any time*. For GNIRS, and any other non-NIRI systems, this number only represents the OIWFS motors. If the system attempts to move a motor when the maximum number of motors is already in motion, then the motion of that motor will silently be delayed until the requested action may be carried out; no error or warning will be issued.
- It is the responsibility of the instrument to combine the value of the wfs:wfsBeam with information from the components controller to produce the value of the global wfsBeam SIR record. This field indicates only that there is no obstruction, not that light is actually getting to the detector. Furthermore, it does not consider the obscuration caused by the pick-off mirror.

2-3.2.3 Detailed SIR record description

2-3.2.3.1 The lockMaxMove, lockActive, and lockMoving records

The lockMaxMove, lockActive, and lockMoving records provide a mechanism to restrict the number of mechanisms which may move simultaneously (necessary, because of power supply restrictions). The lockMaxMove record contains the maximum number of mechanisms which may be moved simultaneously. The lockMoving record contains the number of mechanisms which are currently moving. The lockActive record contains the number of mechanisms which are moving or which are waiting to move.

If an attempt is made to move more mechanisms than are permitted, then the system will silently postpone the motion, so that no more than the maximum number of mechanisms are in motion.

This restriction does not apply to the gimbal (prbx, prby) mechanisms, which are permitted to move at any time.

2-4.0 Software/Control Function Interface (Components Controller)

2-4.1 Overview

2-4.1.1 Communications Infrastructure

The communications infrastructure is the same as described in [16].

2-4.2 Implementation

The instrument may control the temperature interlock system by writing to the lock records which are described above. Alternatively, it may provide a status and information record² which will be used to automatically update the lock records.

2-4.2.1 Monitoring an external record.

The temperature interlock may be set by using a record in another IOC. This feature must be enabled as a configuration option (see [7], section 6.4, “Configuration,” p. 8 and section 7.4, “Configuration,” p. 11). *All three of the variables defined below must be set as described, or all three must be left unset.*

Set the configuration variable *NIRI_TMPVAR* to the name of a field of an EPICS record which contains a numeric value. If the field contains a 0, the value of the temperature interlock will be “CHANGING.”; a 1, the value is “COLD”; and a 2, the value is “WARM”.

The configuration variable *NIRI_TMPSCAN* should be set to the rate at which the field defined by *NIRI_TMPVAR* should be monitored. This should be a valid capfast scan rate, e.g., “1 second”. There is no reason to scan at a rate faster than “1 second.”

The configuration variable *NIRI_TMPHB* should be set to the name of a field which is changing at least as frequently as *NIRI_TMPSCAN*, and which resides within the same IOC as *NIRI_TMPVAR*. This is used to detect communications problems and IOC problems, and

2. Gemini requires a SIR record to be the interface between two systems. The software will work correctly with any record that can return a numeric value.

will automatically force the system into a “CHANGING” (locked) state if there is a problem.

2–4.2.2 Status Information Records

The following tables show the status information provided by the NIRI/GNIRS OIWFS components controller which are of importance to the instrument as a whole. The tables have the following columns:

SIR record. The part of the name of the EPICS SIR record which follows the “niri:” prefix³.

Units. The units in which the quantity is stored

Comments. A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.

TABLE 3. NIRI/GNIRS OIWFS status Information

SIR record (“niri:wfs:” +)	FITS keyword	FITS includ ed?	Type	Units	Comments
lockTmpVal		no	string		Current temperature state (“COLD” “WARM” “CHANG- ING”)
lockTmpHB		no	long		A non-zero value indicates a communications error or a problem with the IOC containing $\{NIRI_TMPVAR\}$ and $\{NIRI_TMPHB\}$

2–5.0 Safety Issues

The instrument software must always call lockTmpSel when the system is expected to be at a non-equilibrium temperature. If the alternative interface is used (see section 2–4.2, “Implementation,” p. 10), the sir records which are being monitored must hold the appropriate temperature state. *Various components may not operate correctly (and could be damaged) if they are moved when there is a significant temperature gradient across the component (more than 50 degrees C).*

2–6.0 Activity Completion

The interlocks only prevent new actions from starting. They do not interrupt currently executing commands.

3. Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI database (for example “niri:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:”.

In order to be sure that the filter wheel and the focusing stage have stopped, it is necessary to monitor their CAR records. These records are standard CAR records; when the mechanism is idle, these records will have the value “IDLE”.

The Gimbal Mirror is unusual; if it is in “follow” mode, it is possible for the mechanism to be in motion, even if the CAR record shows “IDLE”. For this reason, the CAR record should not be used to determine mechanism activity. Instead, use the IsIdle SIR record. When this SIR record has a zero value, the gimbals are guaranteed to be inactive.

TABLE 4. NIRI/GNIRS OIWFS Activity Records

Record ("niri:wfs:" +)	Record Type	Type	Units	Comments
IsIdle	SIR	long 0=TRUE, 1=FALSE		Both axes of gimbal mirror are inactive.
filtC	CAR	Standard CAR values		State of the filter wheel.
focC	CAR	Standard CAR values		State of the focusing stage.